

VIBRATION-BASED PREDICTIVE MAINTENANCE

Continuous diagnosis for Holcim critical assets.

QARTIA Smart Technologies proposes a solution combining sensing, proprietary electronics, industrial communications and artificial intelligence to anticipate failures in plant rotating machinery.

Focus

Industrial vibration and predictive maintenance.

Objective

Fewer unplanned shutdowns and higher availability.

Format

Executive pilot, measurable and scalable by asset.

INDUSTRIAL VIBRATION

From raw data to maintenance action.

QARTIA develops predictive maintenance solutions based on advanced vibration analysis to anticipate failures in rotating machinery and industrial assets.

The technology captures high-resolution signals directly in the plant, continuously analyses asset condition and turns vibration data into actionable alerts for operations, maintenance and engineering teams.

The system detects misalignment, imbalance, bearing wear, lack of lubrication and other degradation patterns early, before they cause unplanned shutdowns.

QARTIA integrates sensing, proprietary electronics, private DECT NR+ networks and Artificial Intelligence models to deploy an end-to-end solution, even in complex industrial environments or where no previous communications infrastructure exists.



Real plant deployment for vibration-based predictive maintenance, industrial connectivity and continuous asset diagnosis.

HOLCIM FIT

Designed for demanding production environments.

In building materials plants, anticipating vibration-related failures helps keep production running, reduce risk and extend the useful life of assets with high mechanical load.

M

Motors, gearboxes and couplings

Monitoring imbalance, looseness, misalignment and bearing degradation.

C

Conveyors, pumps and fans

Alert prioritisation for equipment that affects production, transport and safety.

P

Connected plant

Data available to maintenance, operations and engineering without relying on manual rounds.



Deployment of industrial sensing on critical assets and continuous operation points.



Installation on the asset to capture mechanical condition and detect failure progression.

Fewer shutdowns, higher availability and decisions before failure.

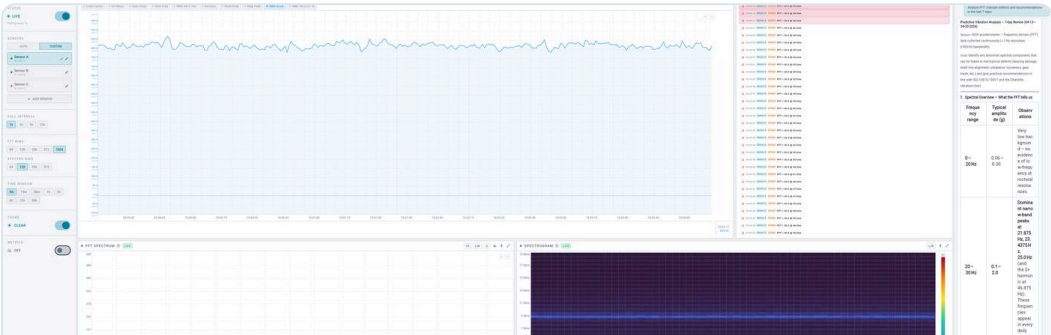
The value of the system is not only measuring vibration, but turning it into a shared view of severity, trend and intervention priority.



12 KHZ

High-resolution sensing

Vibration capture in time and frequency domains to detect real changes in machine condition.



24/7

Continuous diagnosis

Continuous severity and rotating defect type available to act before failure.



ISO 10816 / 20816



VISIBLE ROI

Expert diagnosis, edge and AI-based prediction.

QARTIA's technical architecture is based on proprietary hardware, industrial communications and an expert diagnosis engine with flexible deployment.

- Detection of misalignment, imbalance, bearing wear, lack of lubrication or cavitation.
- Accelerometers up to 12 kHz, acquisition platform and FFT for continuous measurement.
- **Wireless DECT NR+ 5G communications, operator-free, non-cellular, decentralised and in licence-exempt spectrum. Higher Coverage and Mesh Topologies to reach every location across the Cement Plant**
- LLM diagnosis assistant in client, on-premise or cloud environments, with judgement equivalent to a Category II expert.
- Real-time alert service to warn of increased severity, out-of-range trends, abrupt behaviour changes or growing risk in rotating assets.



System electronics focused on acquisition, processing and continuous plant digitisation.



Modules prepared for DECT transmission, deployment on industrial assets and reliable connectivity.

Predict when failure will occur.

AI uses vibration and operating data to estimate asset evolution and anticipate when failure may appear.



Asset behaviour prediction and probabilistic failure reading for motors and machines.

05

Digital Twin
Visualisation of AI results in a Digital Twin of the monitored machine.

01

Asset signal
Vibration, speed, temperature and operating history feed the model.

02

Failure prediction
AI classifies equipment condition and estimates how failure probability evolves.

03

Recommended action
The system prioritises intervention before an unplanned shutdown.

04

Integrated with RAG
The team can ask in natural language about the diagnosis, probable cause, asset history and maintenance recommendations.

Executive pilot in 4 phases.

A scoped pilot validates data quality, coverage, alert severity and economic value before scaling by area or asset family.

1

Asset selection

Operational criticality, failure history, downtime cost and maintenance window.

2

Field deployment

Sensing and network with coverage, mechanical anchoring and signal quality validation.

3

Diagnosis and AI

Impact report, actionable alerts and recommendations prioritised by asset.

4

Implementation plan

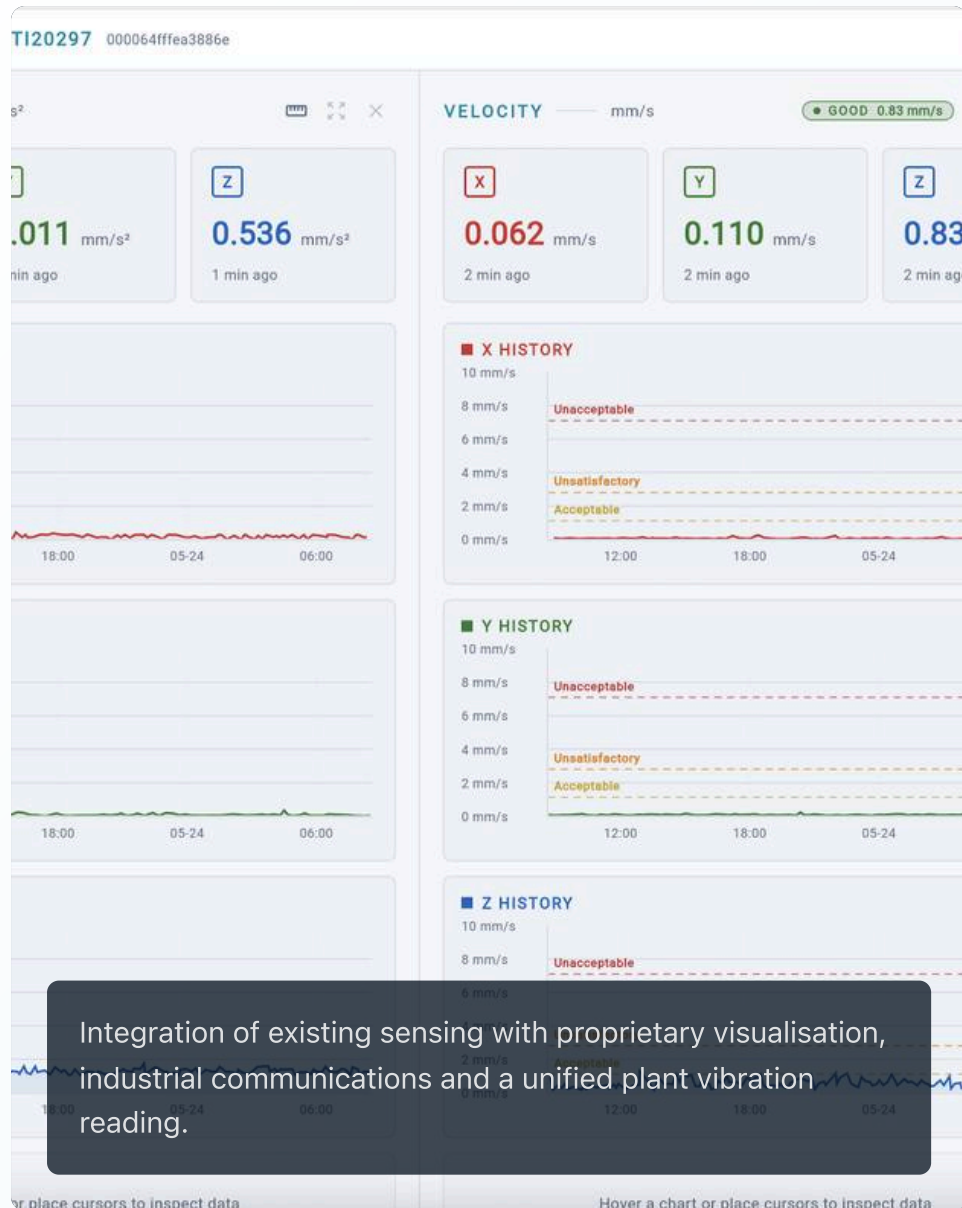
Final economic proposal, scale-up plan and service agreement.

MTBF

Less frequent failures and greater stability in critical rotating machinery.

OEE

Higher plant availability and more productive operating time.



Integration of existing sensing with proprietary visualisation, industrial communications and a unified plant vibration reading.

TECHNOLOGY INTEGRATION

Also if Holcim already has sensing in place.

QARTIA can integrate with its own hardware or with existing factory sensing, avoiding duplication and leveraging previous investments.

The solution is platform- and vendor-agnostic. It can work with existing equipment and technologies such as LoRa, Ethernet or Modbus, adapting to the specific requirements of each industrial environment.

The goal is to deliver a common layer for visualisation, communications and diagnosis so that maintenance, operations and engineering teams read mechanical risk with the same criteria.

This approach makes it easier to start with critical assets and move towards a more autonomous, traceable maintenance model focused on avoided cost.

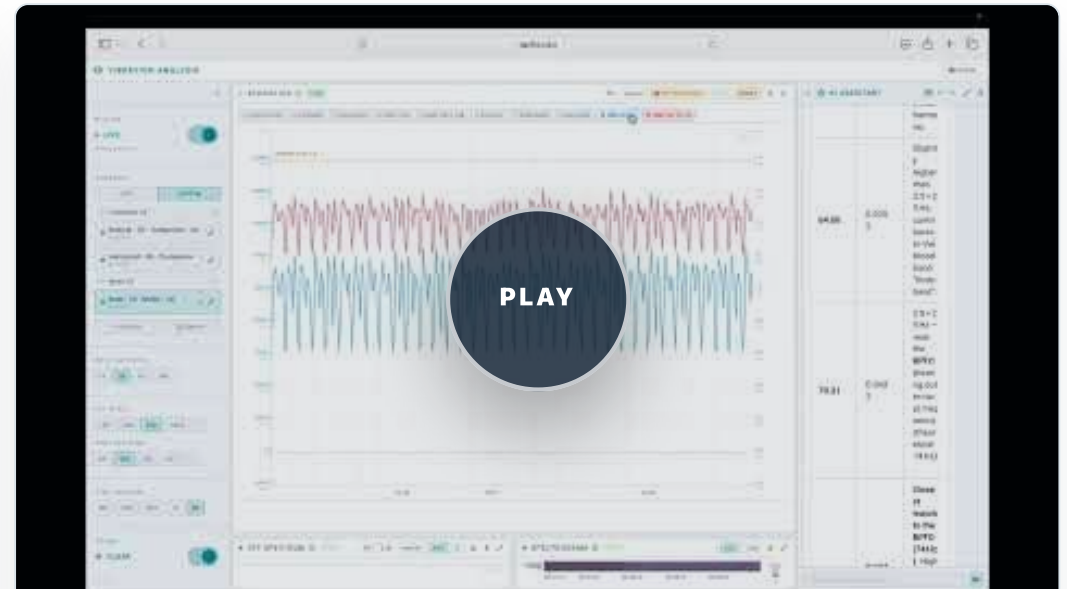
Predictive maintenance and applied RAG.



Plant vibration diagnosis

Video on predictive maintenance based on vibration analysis, focused on deploying the solution on real industrial assets.

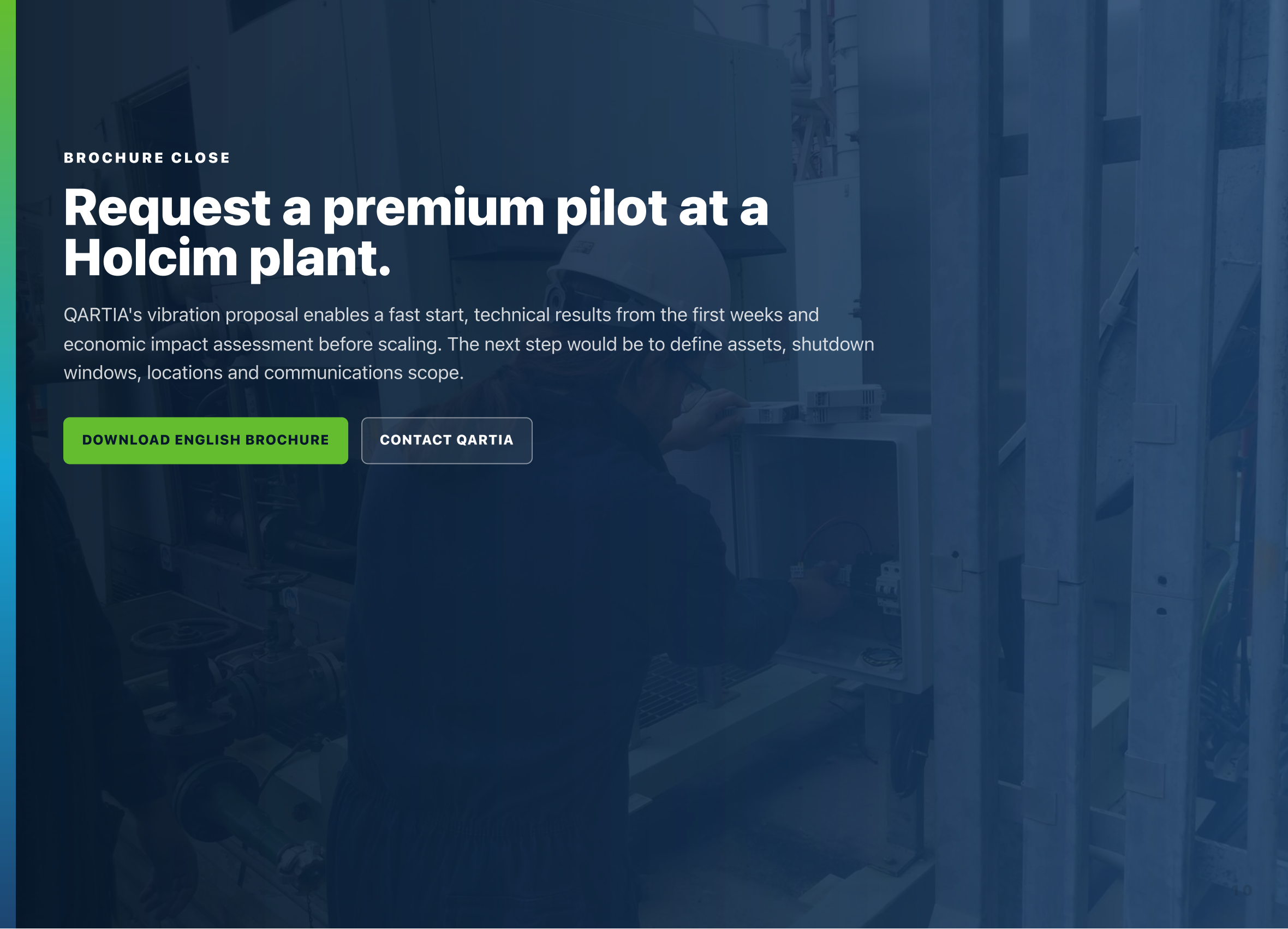
[OPEN VIBRATION VIDEO >](#)



RAG - Operational Chat Bot

Predictive Maintenance Assistant equivalent to a Category II vibration expert.

[OPEN RAG VIDEO >](#)



BROCHURE CLOSE

Request a premium pilot at a Holcim plant.

QARTIA's vibration proposal enables a fast start, technical results from the first weeks and economic impact assessment before scaling. The next step would be to define assets, shutdown windows, locations and communications scope.

DOWNLOAD ENGLISH BROCHURE

CONTACT QARTIA